
ASEN 5335: Aerospace Environment

Module 3: The Ionosphere

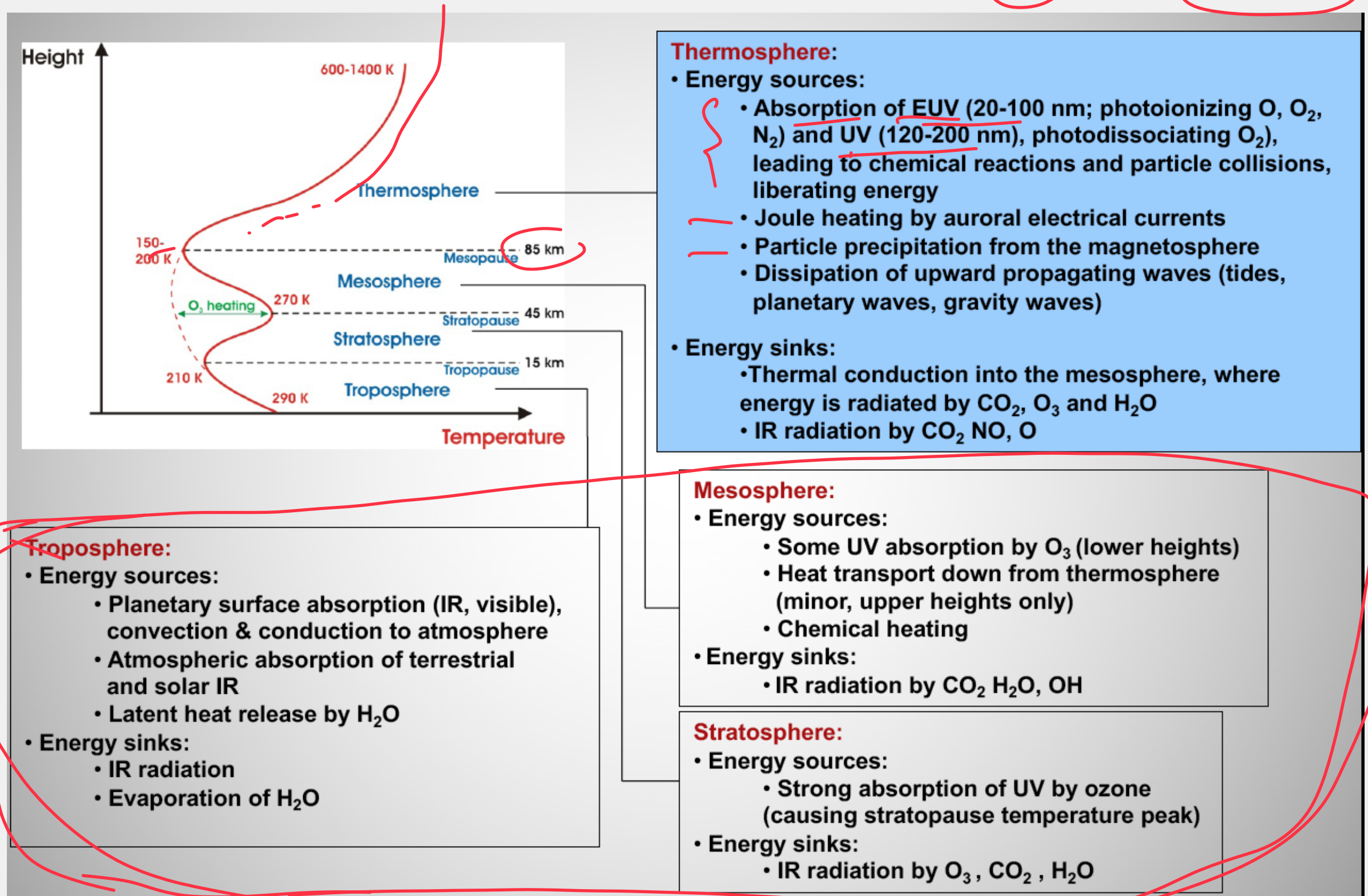
Xinlin Li

Goals of This Module

- ❖ **Understand basic physics of the Earth's ionosphere**
 - ❖ Origin, composition, layers
 - ❖ Variations: diurnal, seasonal, solar cycle, plus other anomalies
 - ❖ Basics of the plasmasphere
- ❖ **Understand effects of the Ionosphere on Spacecraft and technology**
 - ❖ Spacecraft charging
 - ❖ Radio communications and GPS
- ❖ **Learn about Ionospheric space missions, including ICON and AMPERE**

Origin of the Ionosphere

- ❖ The Ionosphere is a product of two regions we've covered already: the **Sun**, and the **Atmosphere**



Absorption in Earth's Atmosphere

$$I = \text{solar flux} = \left[\frac{\text{J}}{\text{cm}^2/\text{sec}} \right] = \left[\frac{\text{W}}{\text{cm}^2} \right]$$

$$dI \sim I dh$$

$$dI = -\sigma n I dh$$

number density: cm^{-3}
absorption cross section cm^2

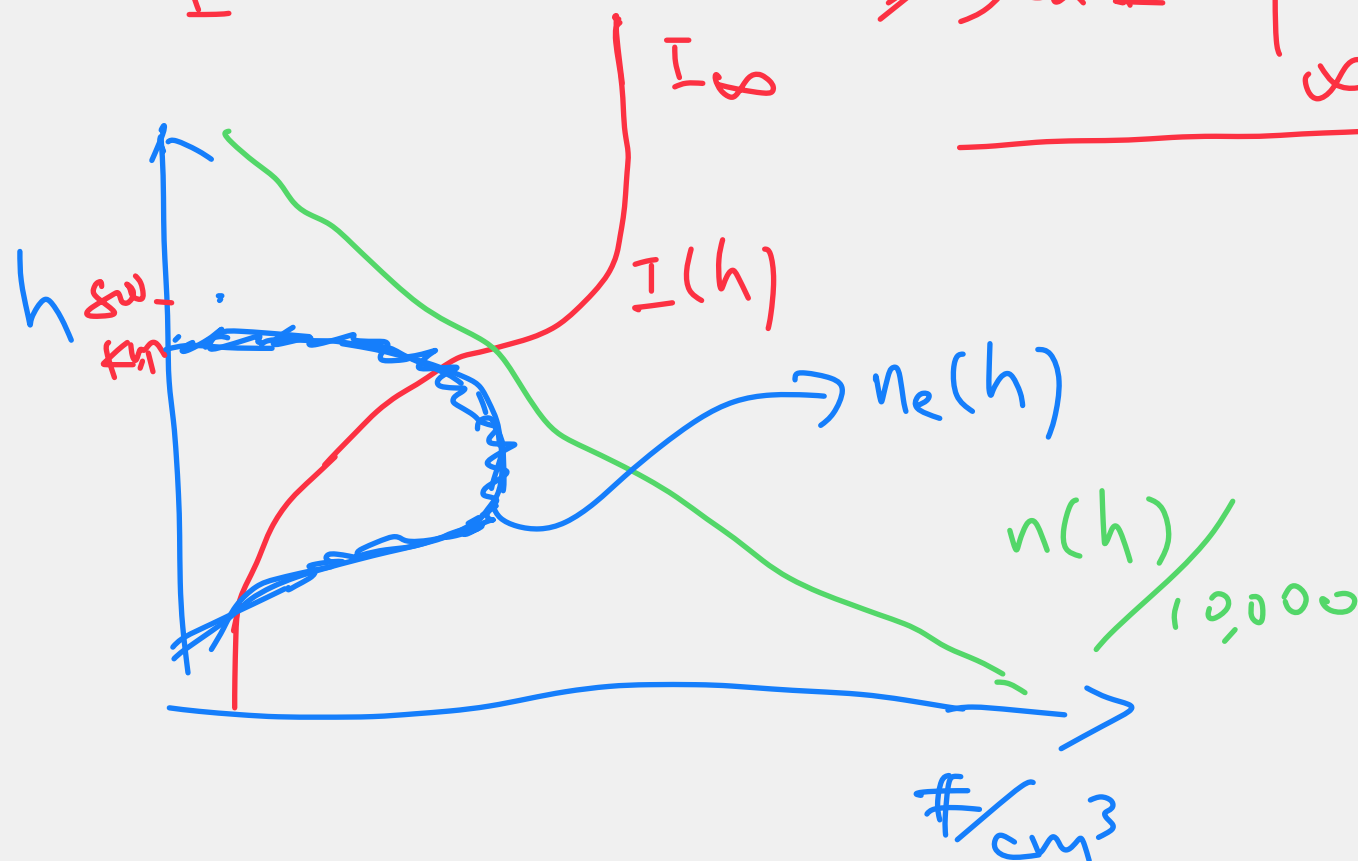


molecule, atom. dependent, λ -dependent

$$\frac{dI}{I} = -\sigma n dh \Rightarrow \ln I \Big|_{\infty}^h = -\int_{\infty}^h \sigma(h) n(h) dh$$

$$I(h) = I_{\infty} e^{-\tau}$$

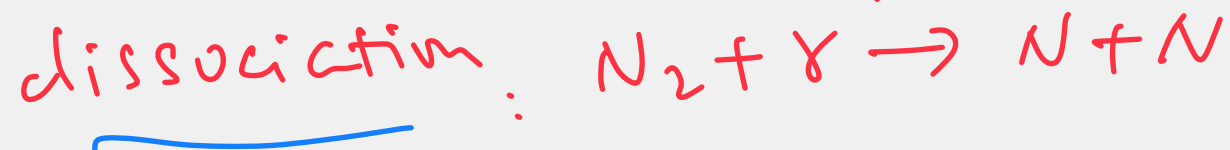
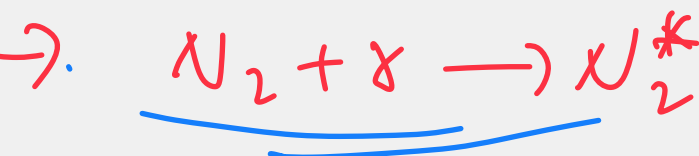
τ : optical length



Atmospheric Heating

- ❖ Though less than 1.5% of TSI, wavelengths < 300 nm are primary source of atmospheric heating from 15–500 km.

- ❖ Excitation; dissociation; ionization
- ❖ Relaxation, association, recombination → heat

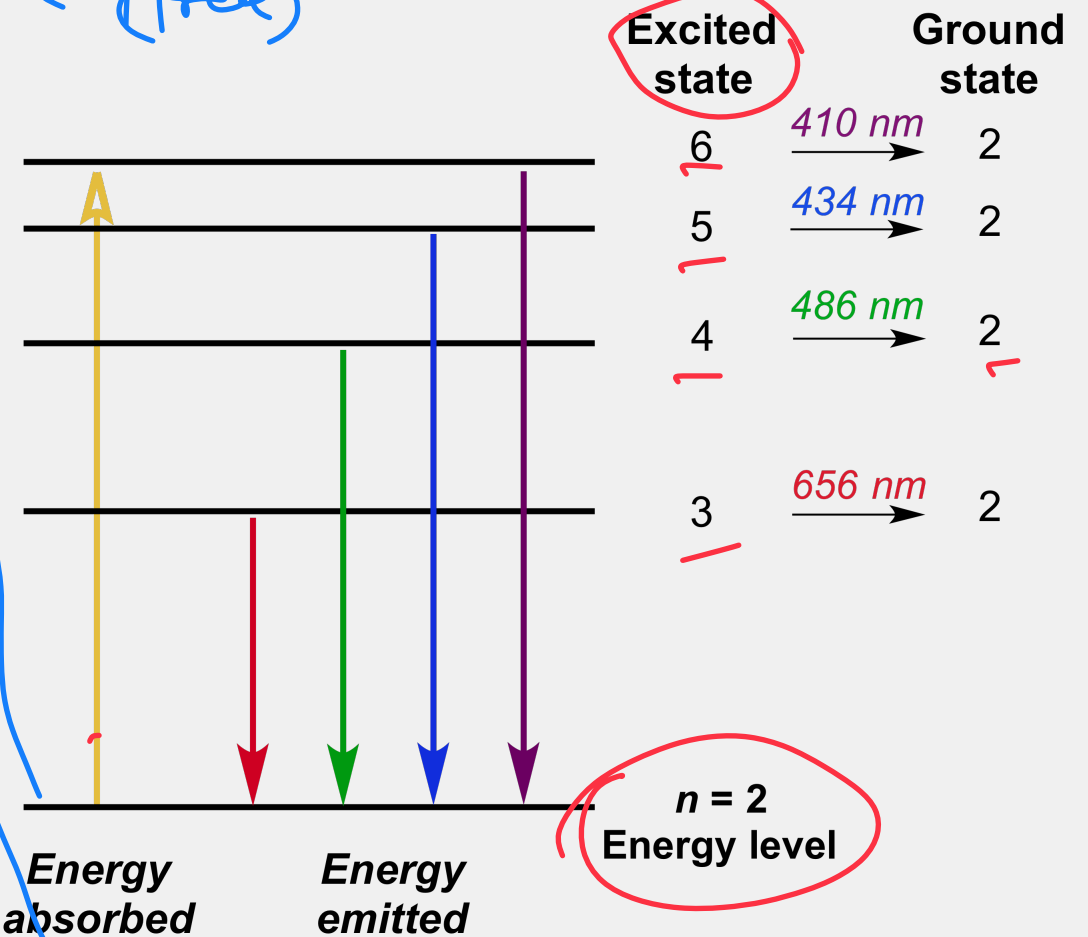


Total Solar Irradiance

$$1361 \frac{W}{m^2}$$

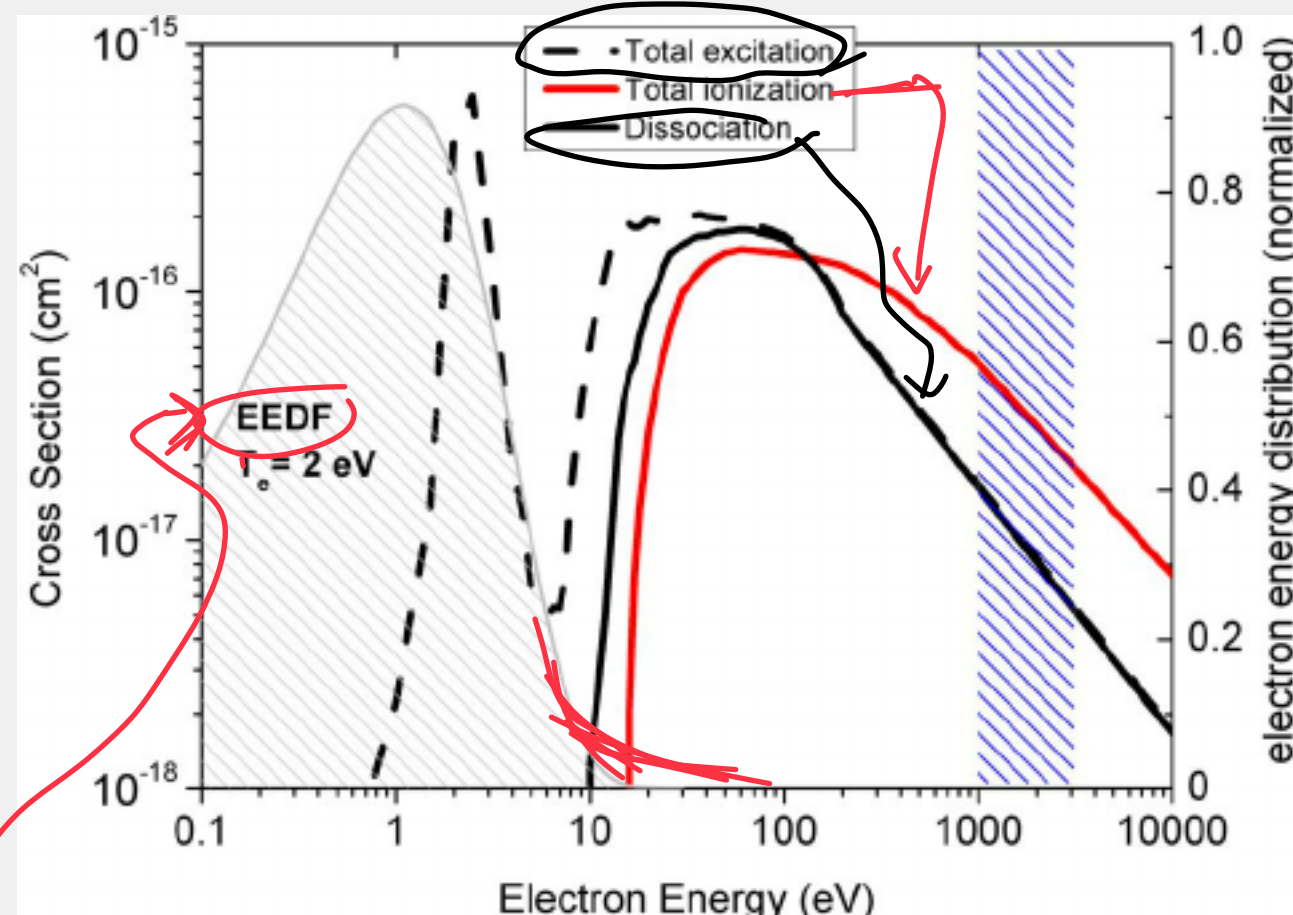
at 1 Au

e^- (free)

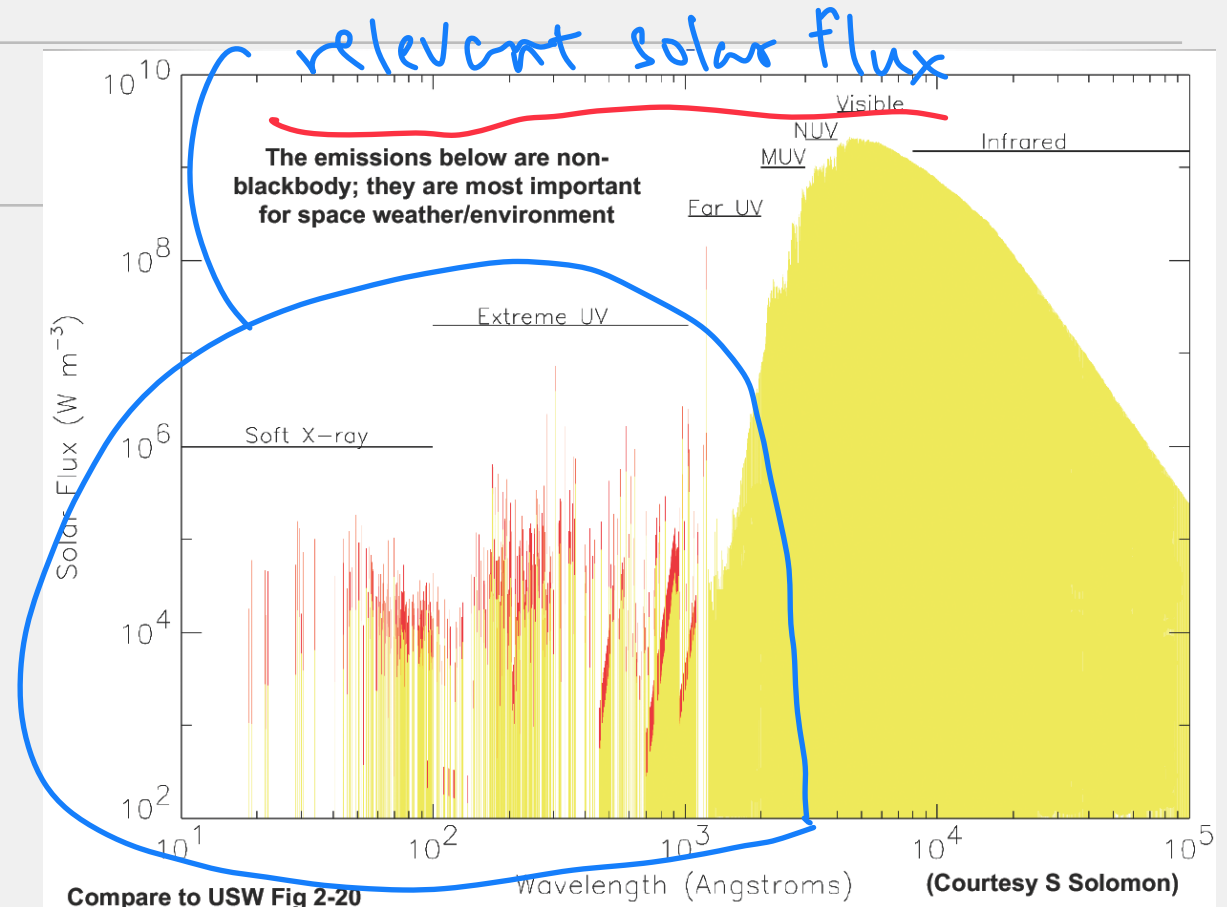


Ionization thresholds

- ❖ Require a minimum energy to free an electron from an atom or molecule
- ❖ Require a photon with at least this minimum energy: "**ionizing radiation**" or sometimes just "**radiation**"
- ❖ **Ionization cross section** provides energy-dependent picture of ionization probability



Electron Energy Distribution Function

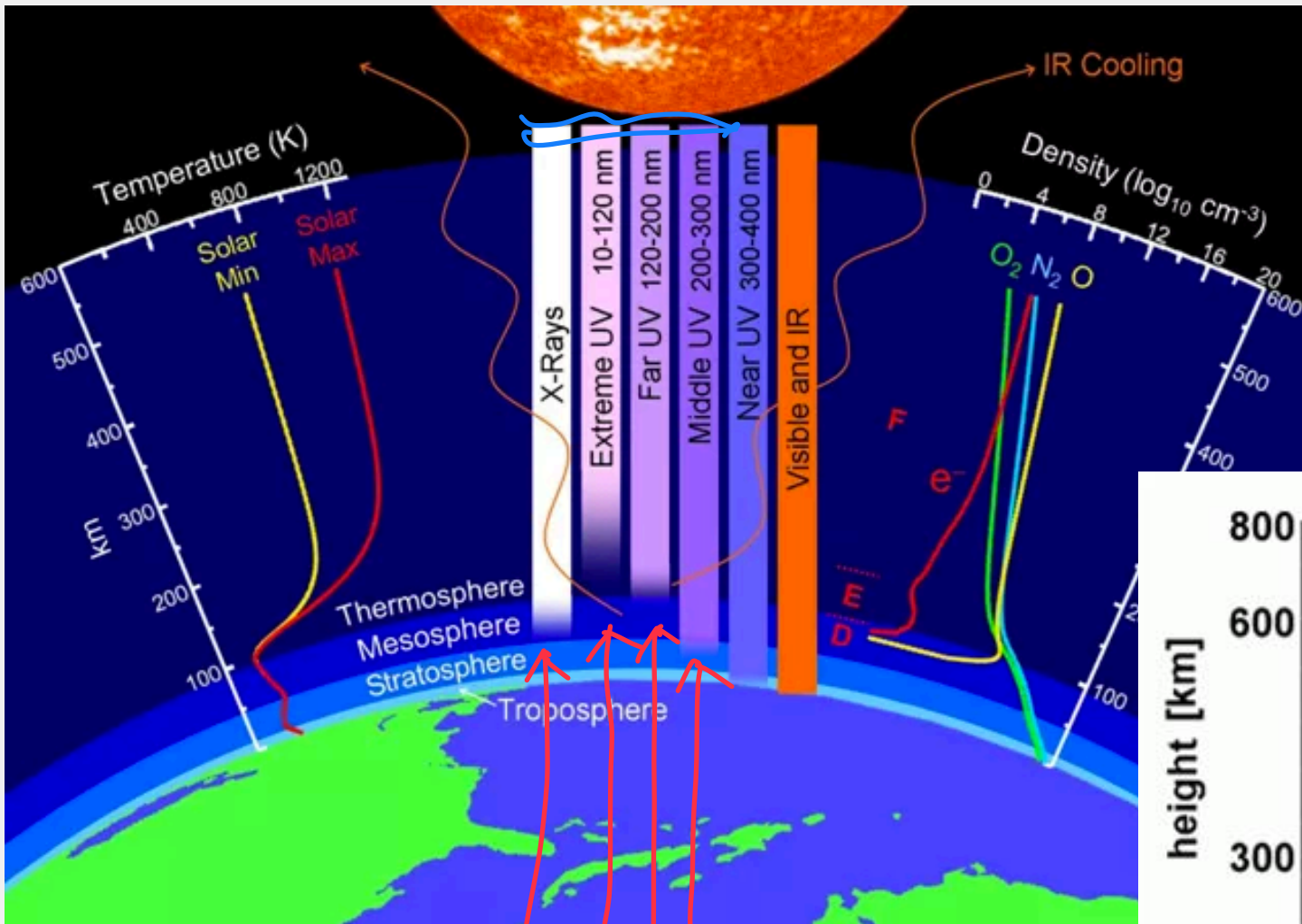


	<i>min</i> Ionization Energy	<i>max</i> Minimum photon wavelength
H	13.6 eV	91 nm (910 A)
He	24.6 eV	50 nm
O	13.62 eV	91 nm
Ar	15.76 eV	79 nm
N ₂	15.6 eV	80 nm
O ₂	12.1 eV	103 nm

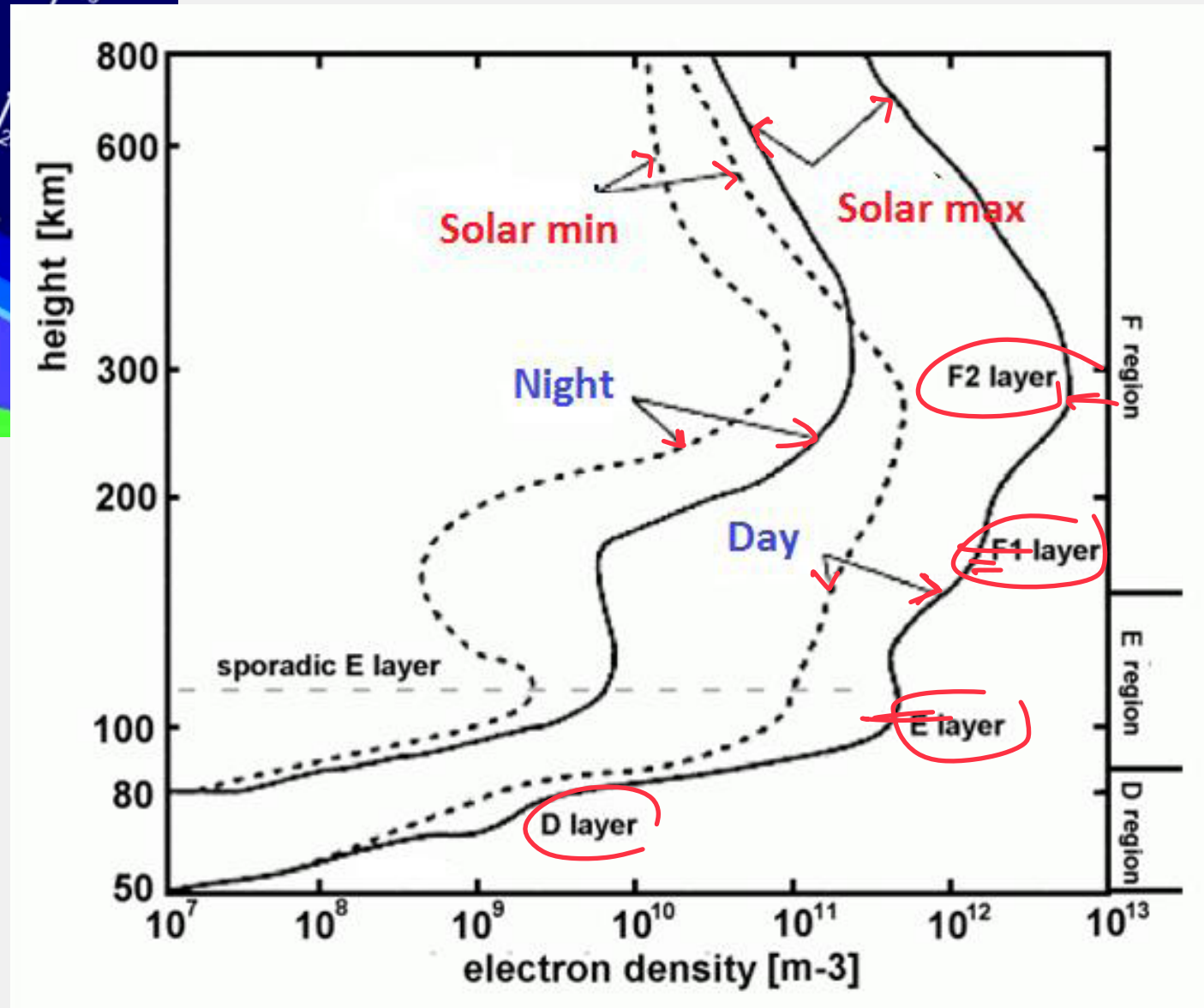
Do we concern about the solar flux in visible and infrared range? why?

Earth's Ionosphere

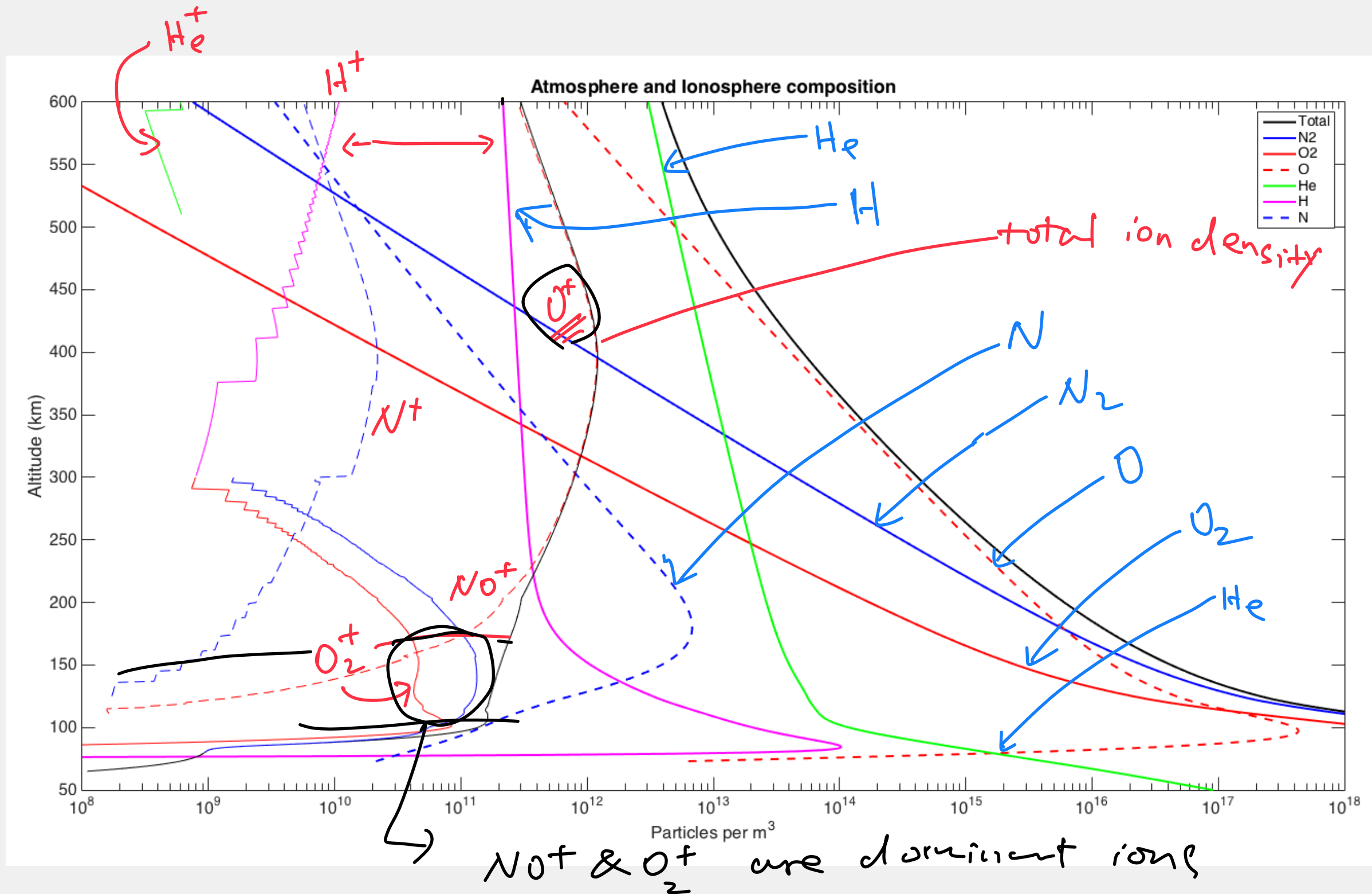
- ❖ Ionosphere altitudes and layers have a lot to do with where solar radiation is absorbed!



relevant
solar flux



Ionosphere Composition and Density



Why is the ionosphere finite in altitude?

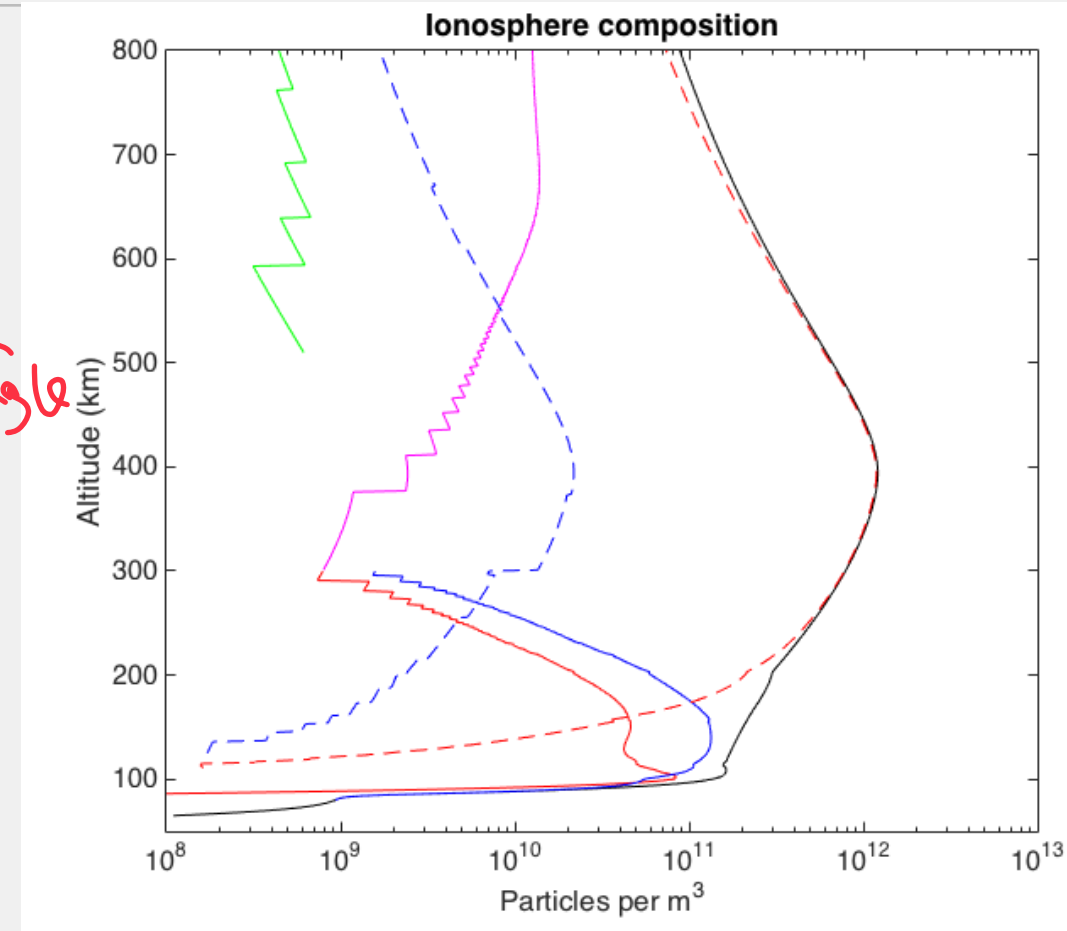
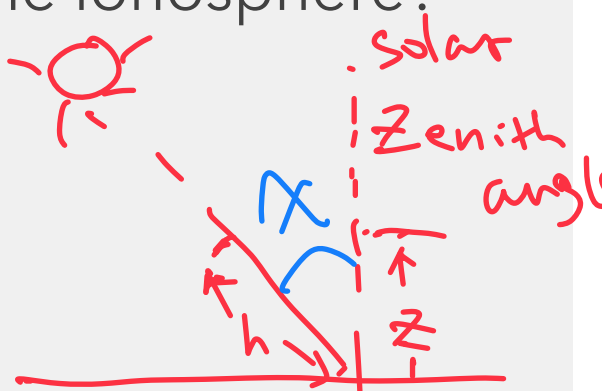
- ❖ The atmosphere is exponentially increasing all the way to the ground. What about the ionosphere?

$$dI = -\sigma n(h) I dh$$

$$= -\sigma n(h) I dz \sec \chi$$

$$I(z) = I_{\infty} e^{-\int_{\infty}^z n(z) \sigma \sec \chi dz}$$

if $n(z) = n_0 e^{-z/H}$



$$I(z) = I_{\infty} e^{-H n(z) \sigma \sec \chi}$$

$$I(z, \lambda, \chi) = I_{\infty}(\lambda) \exp\left[-\int_{\infty}^z \sum_i n_i(z) \sigma_i(\lambda) \sec \chi \cdot dz\right]$$

$$= I_{\infty}(\lambda) e^{-\tau(z, \lambda, \chi)}$$

The relevant solar flux (in UV or shorter wavelengths) will become zero somewhere above the ground, why?

Chapman Layer

Ionization Production Rate : $P - \text{pair}/\text{m}^3/\text{s}$

$$P = I(z, \lambda, X) \cdot n(z) \cdot \underbrace{\sigma \cdot \eta_i}_{\text{ionization efficiency: } 0-1}$$

$$P = I_\infty e^{-H n(z) \sigma \sec X} \cdot (\sigma \eta_i) \cdot n(z)$$

$\sigma_i (\text{cm}^2)$ ionization cross section

$$P = I_\infty \sigma \eta_i n_0 e^{-\frac{z}{H}} \exp(-H \sigma \sec X n_0 e^{-\frac{z}{H}})$$

set: $\frac{dP}{dz} = 0 \Rightarrow z_{\max}$, plug in $z_{\max}$

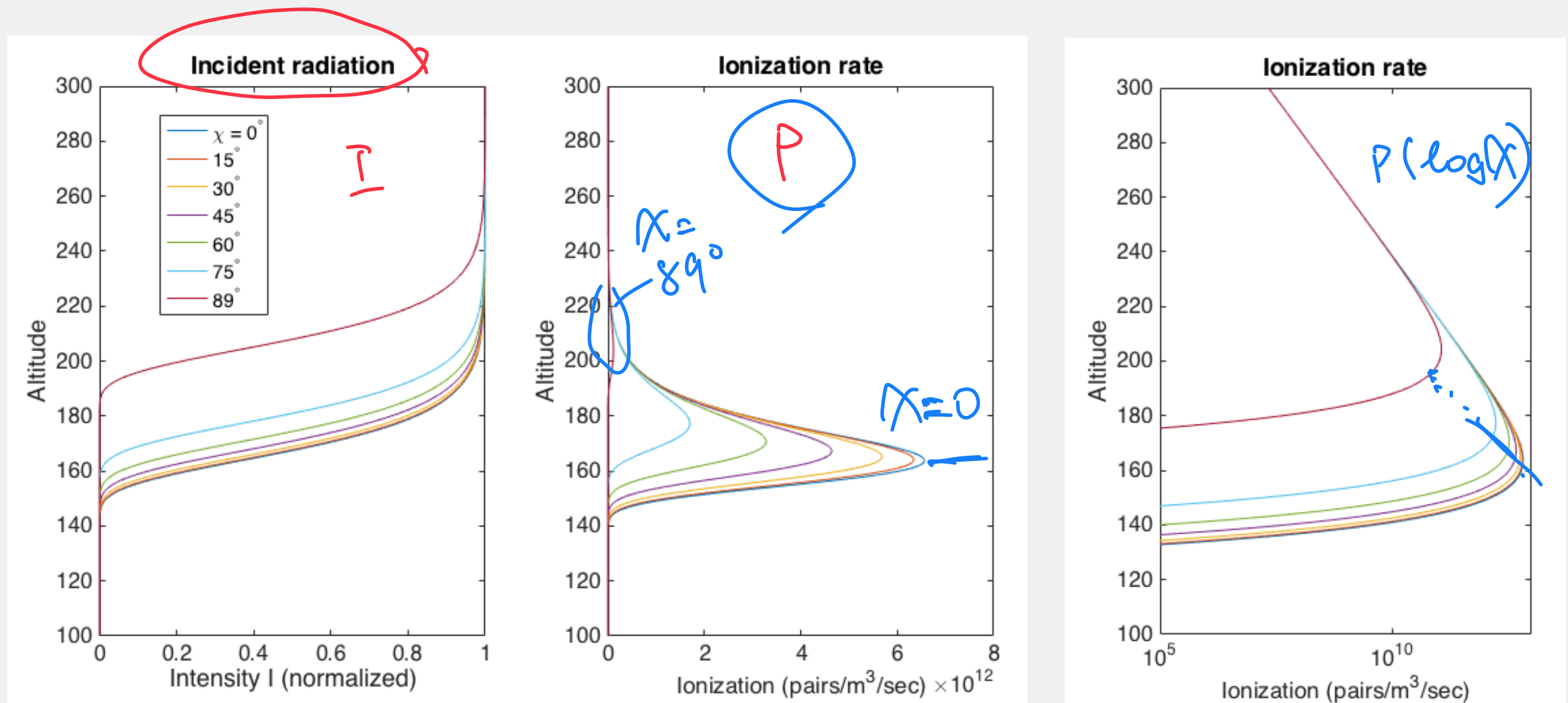
$\Rightarrow P_{\max}$

where, in altitude, will be the max production rate? How to figure this out?

Chapman Layer

- ❖ Production higher for lower zenith angle (i.e. noon)
- ❖ Peak in production is near where intensity is about **half** the incident value

$$Z_{\max} = H \ln(n_0 \sigma H \sec \chi) \quad , \quad P_{\max} = \eta_i \frac{I_\infty}{H} \cos \chi e^{-1}$$



Ionization Chemistry

- ❖ Ionosphere is in equilibrium when ionization production and loss mechanisms balance
- ❖ Production: photoionization; energetic particle precipitation; collisions
- ❖ Loss: recombination; charge exchange; chemistry

$$\frac{dn_e}{dt} = P - L \quad \text{loss}$$

at equilibrium: $\frac{dn_e}{dt} = 0 \Rightarrow \underline{P = L}$

$$L = \alpha n_o n_e \sim \alpha n_e^2$$

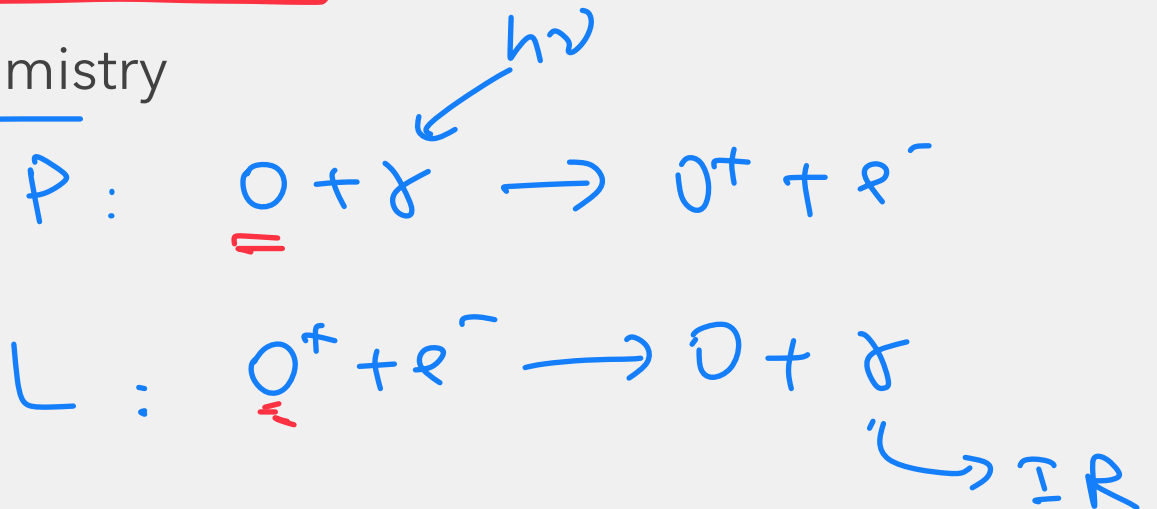
$\uparrow$ recombination coefficient

$$P = L \Rightarrow P = \alpha n_e^2$$

$\Rightarrow$

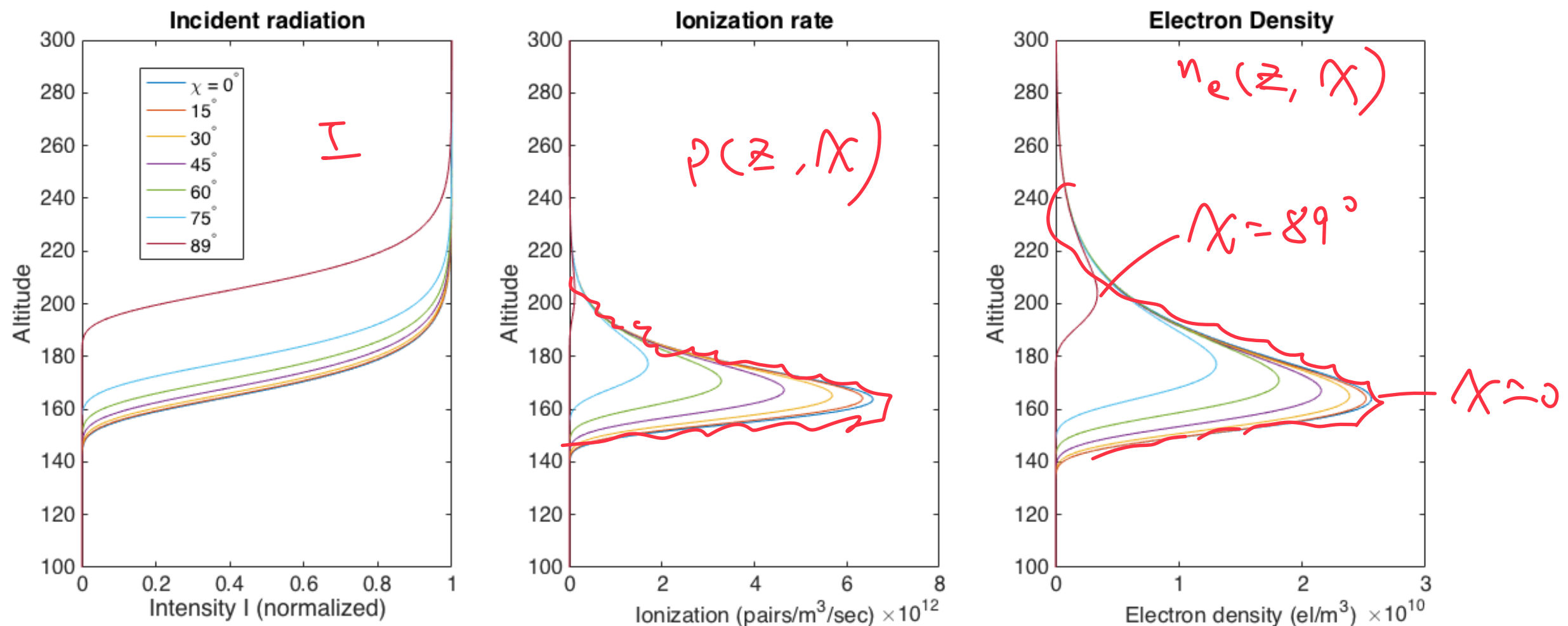
$$n_e = \sqrt{\frac{P}{\alpha}}$$

HW3



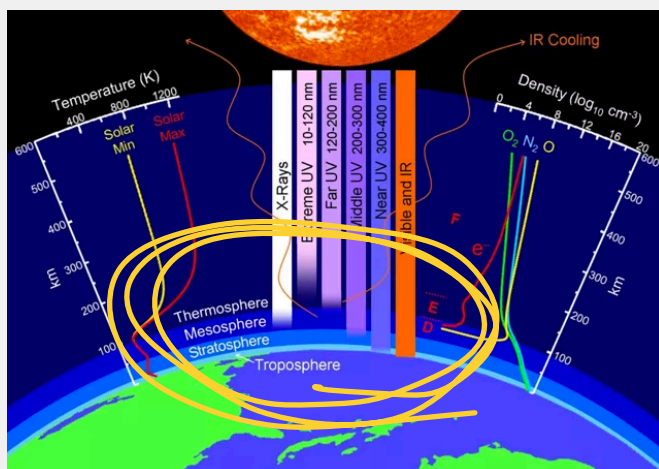
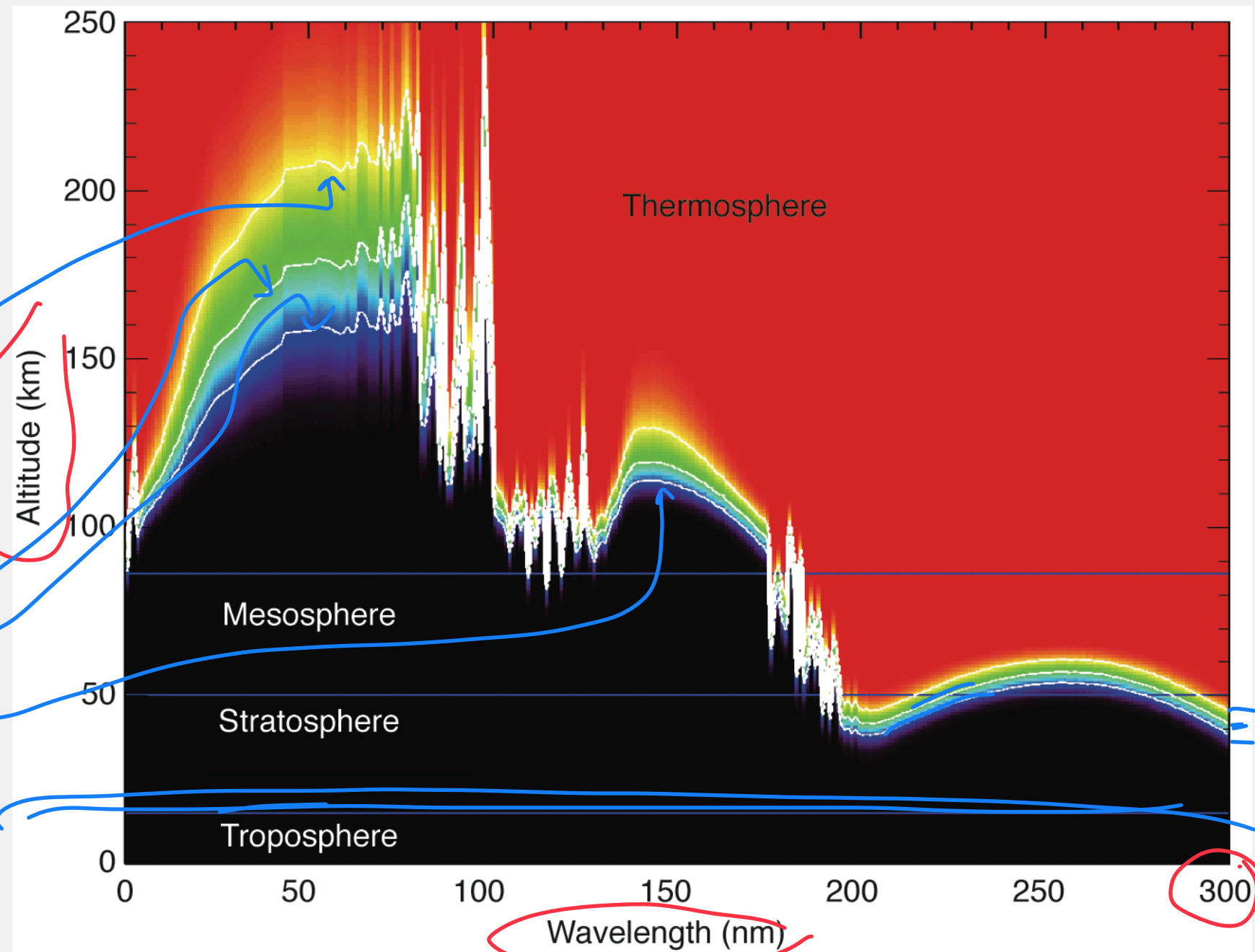
Electron Density Profile

- ❖ Balancing production (ionization) with loss (recombination), we get an equilibrium electron (or ion) density below
- ❖ Higher, less dense for increasing zenith angle
- ❖ Reminder: this is for a single species, and single photon wavelength!



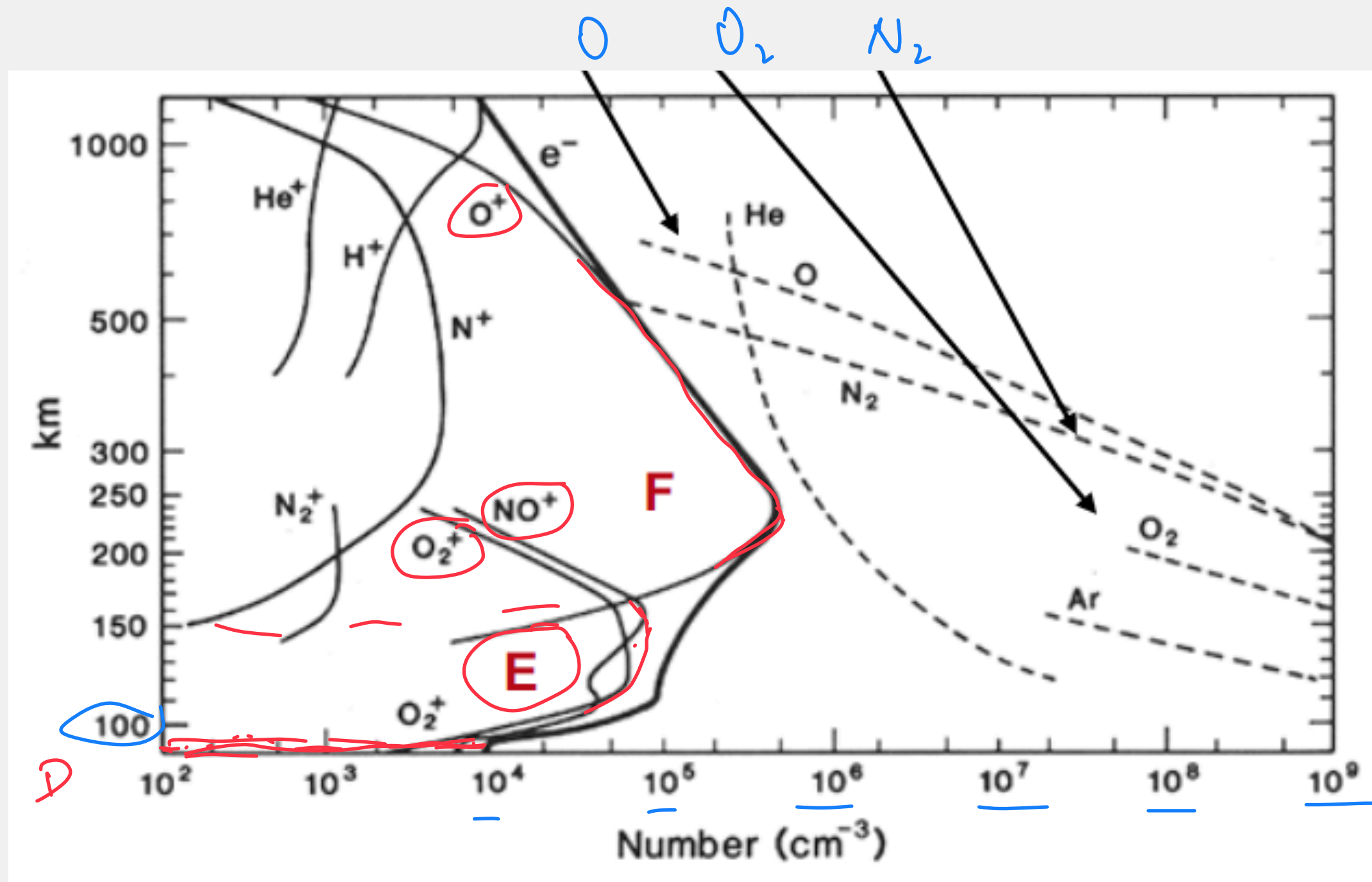
Ionosphere Layers

- ❖ Different wavelengths are absorbed at different altitudes, by different species
- ❖ $I(z)$ depends on wavelength-dependent absorption for each species
- ❖ $P(z)$ depends on wavelength-dependent ionization cross sections for each species
- ❖ Right: top white curve is $I(z)$ decay by $e^{-0.5}$; middle white curve by e^{-1} ; bottom white curve by $e^{-1.5}$
- ❖ Red areas: $I(z)$ is basically I_∞
Black areas: $I(z)$ is basically zero



Another way of looking at it

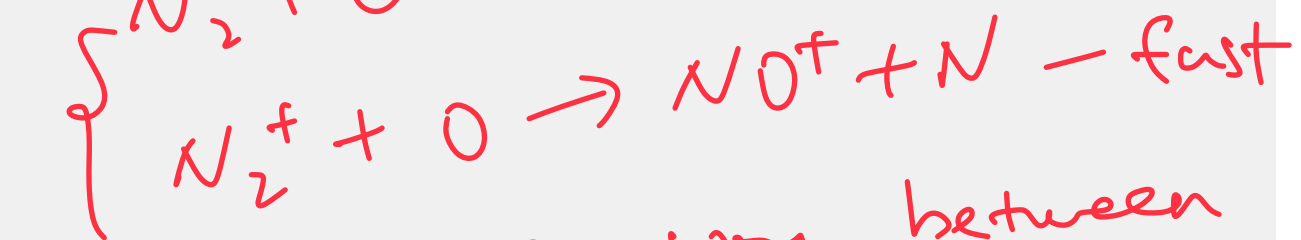
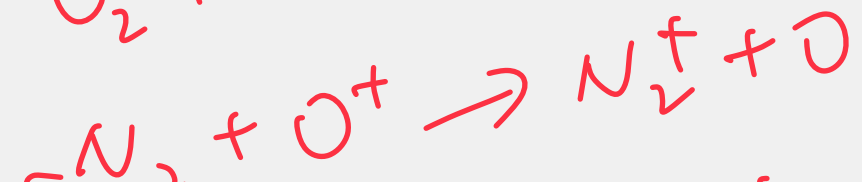
- ❖ Ionosphere layers depend on composition (atmosphere) and radiation (solar spectrum and flux)
- ❖ D, E, and F regions are dominated by different ions, depending on the neutral species



Primary Production / Loss channels



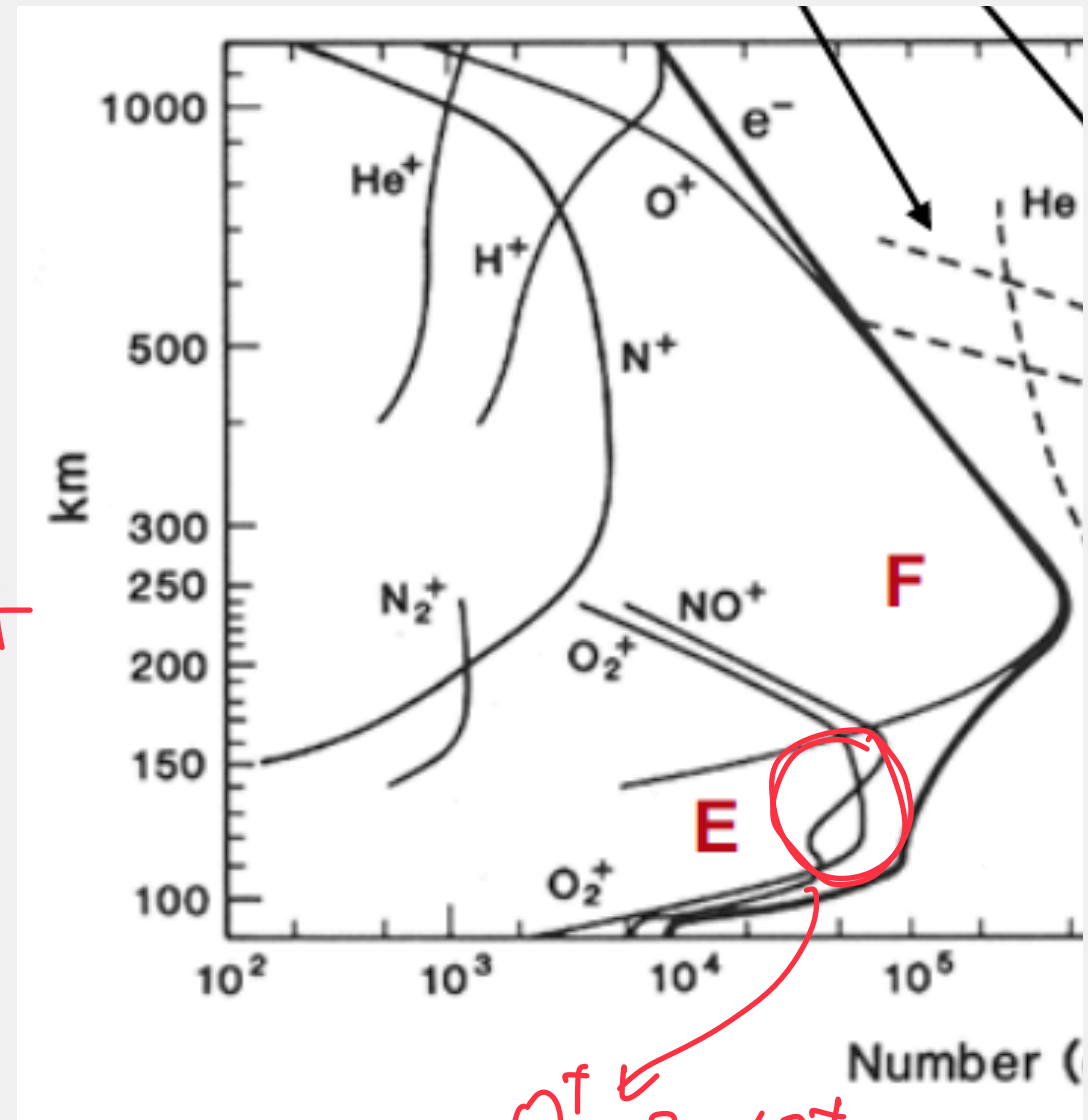
charge exchange :



Chemical reaction between O^+ & $N_2 \rightarrow NO^+ + N$



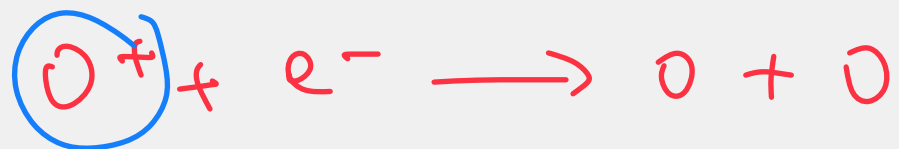
Recombination:



O_2 ✓ exot

E-region

Below $\sim 120 \text{ km}$, $[O_2] \gg [O]$, E-region dominated by



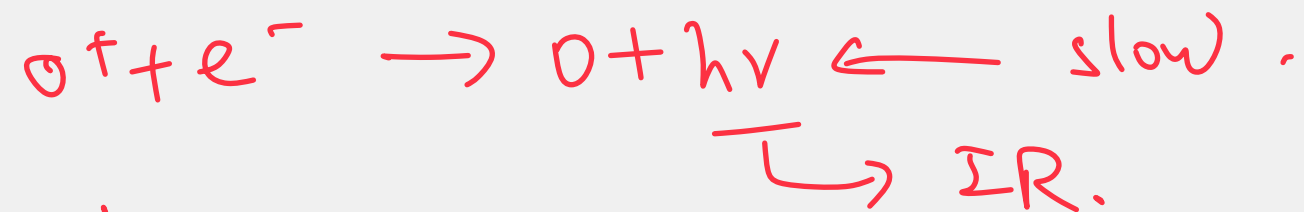
$$L = \alpha n_{O^+} n_e \propto n_e^2, \text{ if } P = L$$

& also Not

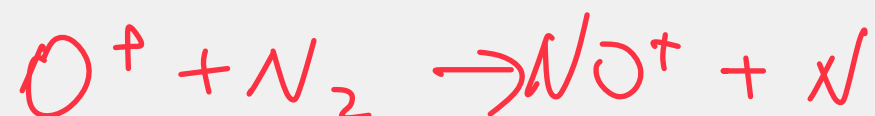
because abundant N_2

$$n_e = \sqrt{\frac{P}{\alpha}}$$

F-region - F1



instead :

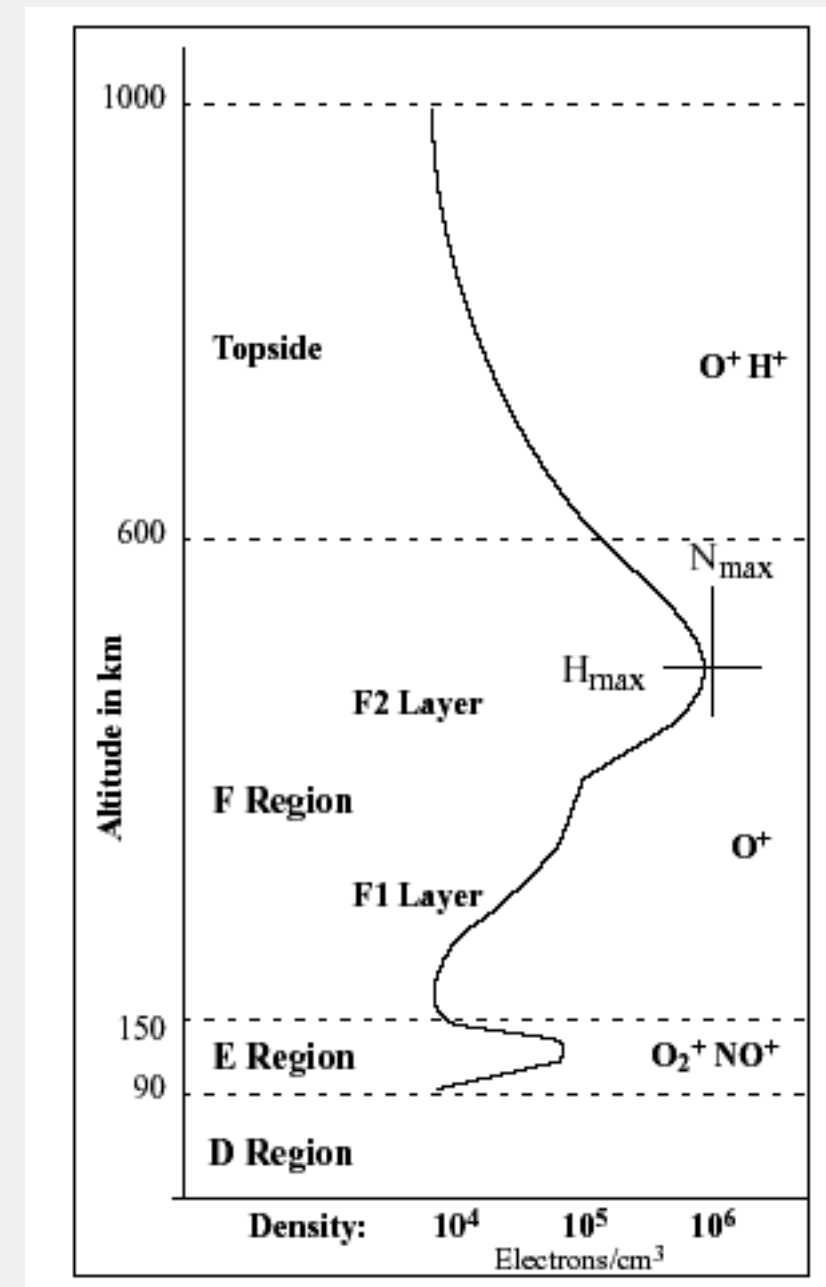


then:



$$L = \alpha' n_{O^+} n_e \propto \alpha n_e^2$$

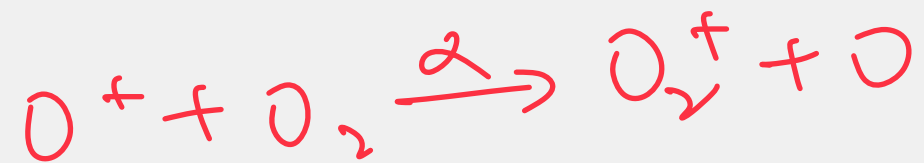
Chapman layer



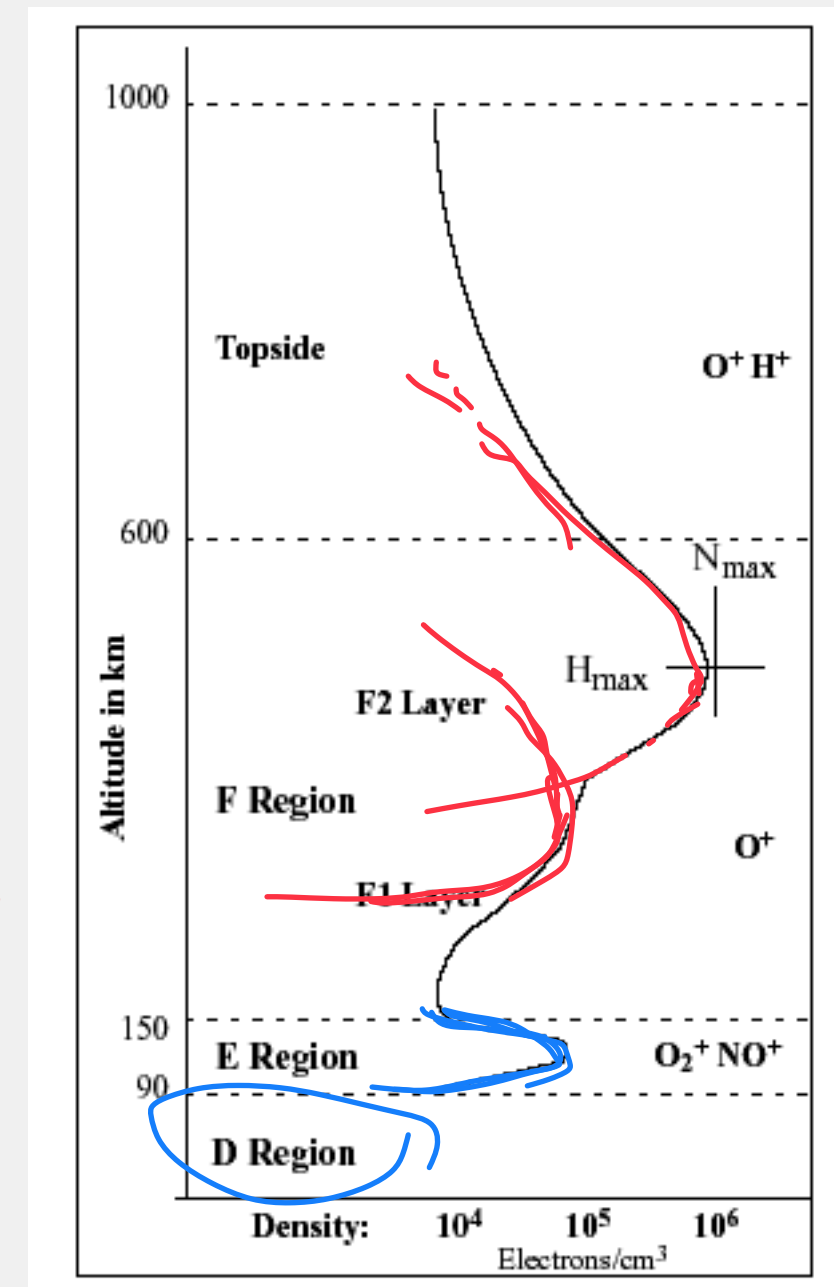
F-region - F_2



charge exchange



$$\text{loss} \sim \alpha^2 n_{O^+} n_{O_2} \sim \alpha n_{O_2^+} n_e \quad \text{with } n_e^2 \text{ circled}$$



D-region

- ❖ D-region is known for low electron densities, light and heavy positive and negative ions, and complex chemistry. The neutral density is much higher there!

- ❖ Production:

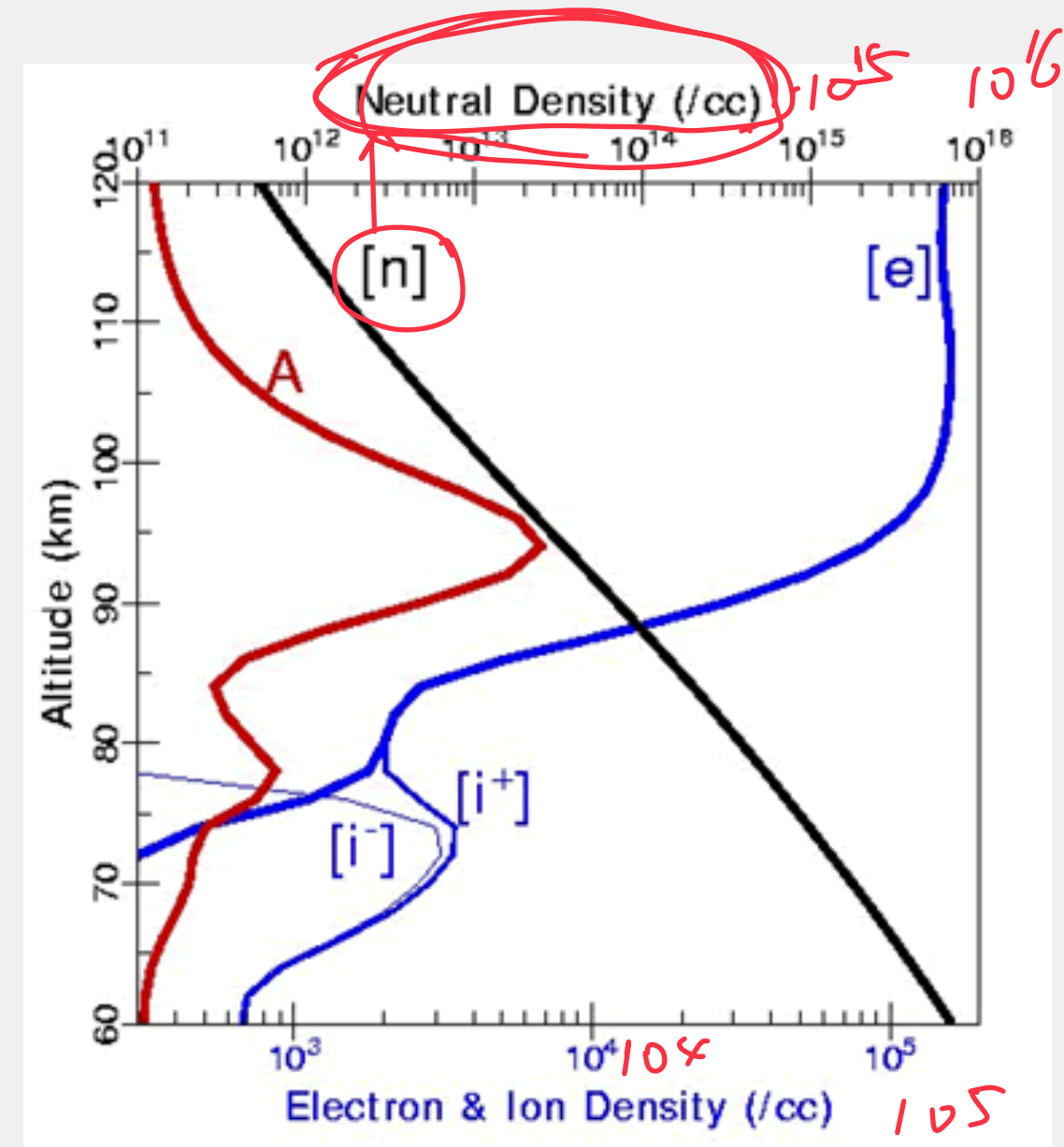
- ❖ $N_2 + \gamma \rightarrow N_2^+ + e^-$
- ❖ $O_2 + \gamma \rightarrow O_2^+ + e^-$
- ❖ $NO + \gamma \rightarrow NO^+ + e^-$

- ❖ Negative ions formed by attachment processes:

- ❖ $O_2 + e^- + O_2 \rightarrow O_2^- + O_2$
- ❖ $O_2 + e^- \rightarrow O_2^- + \gamma$

- ❖ Detachment:

- ❖ $O_2^- + \gamma \rightarrow O_2 + e^-$
- ❖ $O_2^- + O_2 \rightarrow O_2 + e^- + O_2$

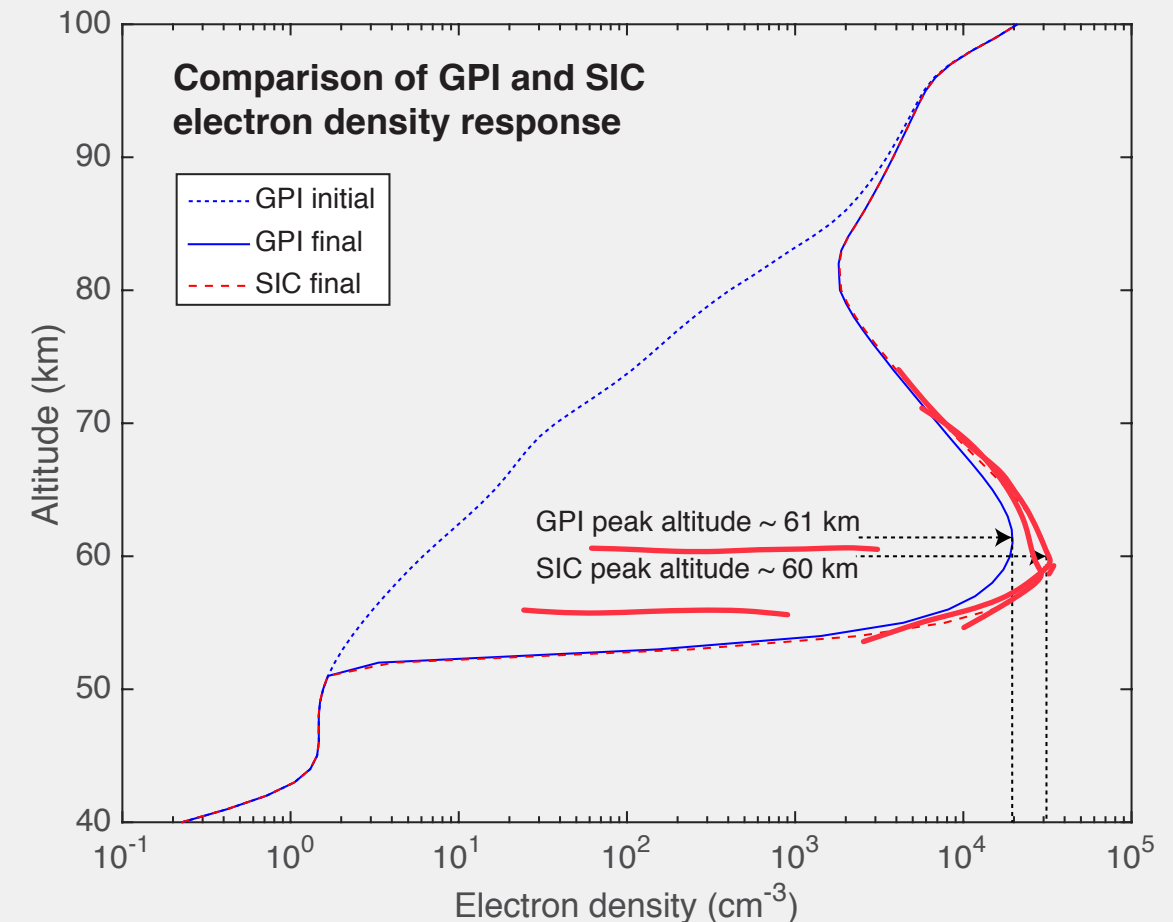


D-region chemistry and "cluster ions"

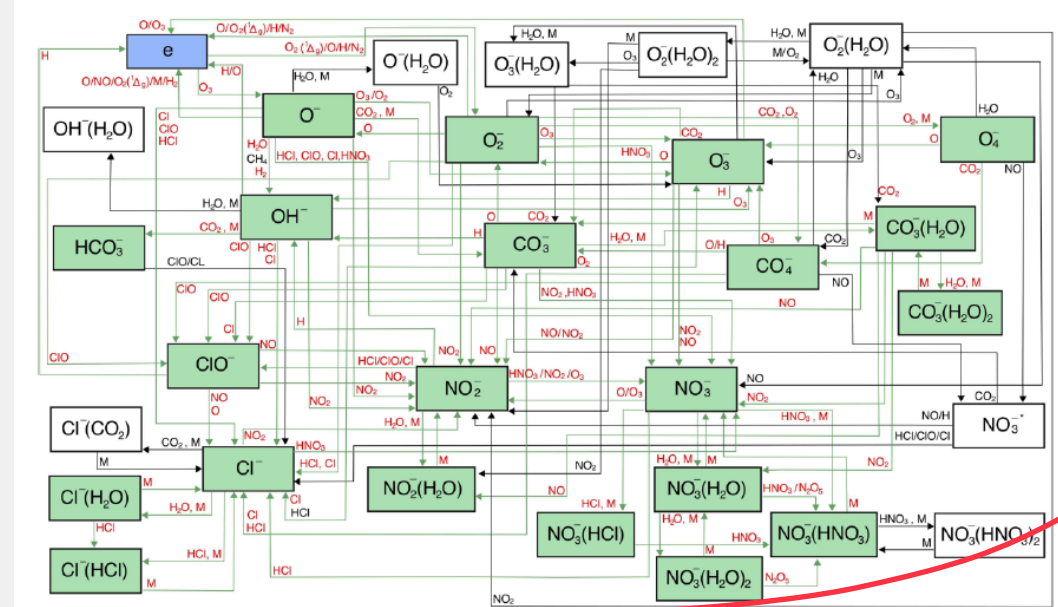
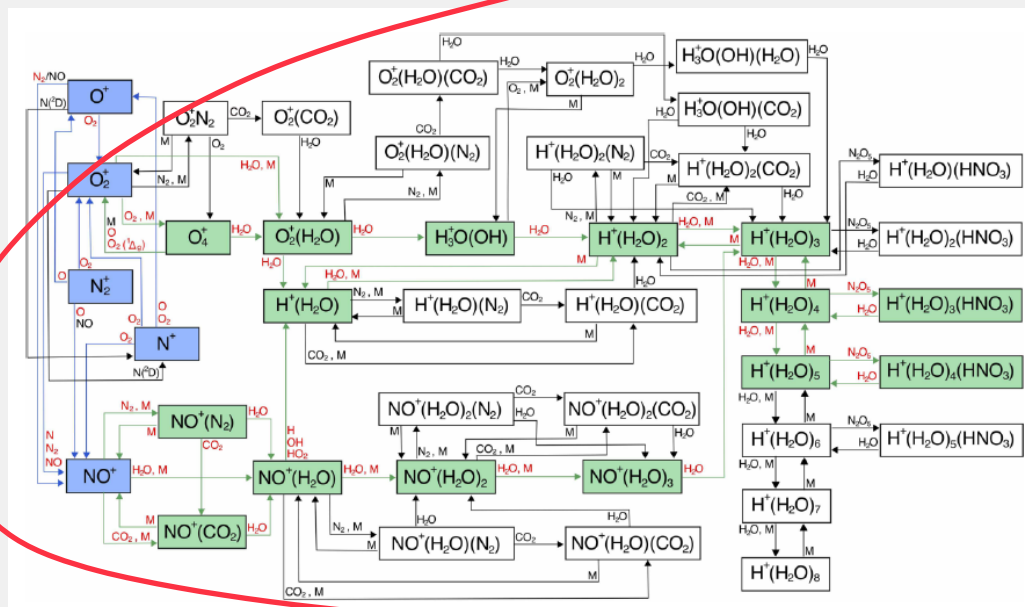
- ❖ D-region also contains heavy water cluster ions of the form $(\text{H}_2\text{O})_n\text{H}^+$
- ❖ Requires more complex chemistry models to evaluate n_e profiles

Glukhov Pasko Inan (GPI) 2009 model:

$$\begin{aligned}\frac{dN_e}{dt} &= \beta N_e + \gamma N^- + \gamma_x N_x^- - (\alpha_d N^+ + \alpha_d^c N_x^+) N_e \\ \frac{dN^-}{dt} &= \beta N_e - \gamma N^- - \alpha_i(N^+ + N_x^+) N^- - A N^- \\ \frac{dN_x^-}{dt} &= -\gamma_x N_x^- - \alpha_i(N^+ + N_x^+) N_x^- + A N^- \\ \frac{dN^+}{dt} &= \alpha_d N_e N^+ - \alpha_i(N^- + N_x^-) N^+ - B N^+ \\ \frac{dN_x^+}{dt} &= \alpha_d^c N_e N_x^+ - \alpha_i(N^- + N_x^-) N_x^+ + B N^+\end{aligned}$$



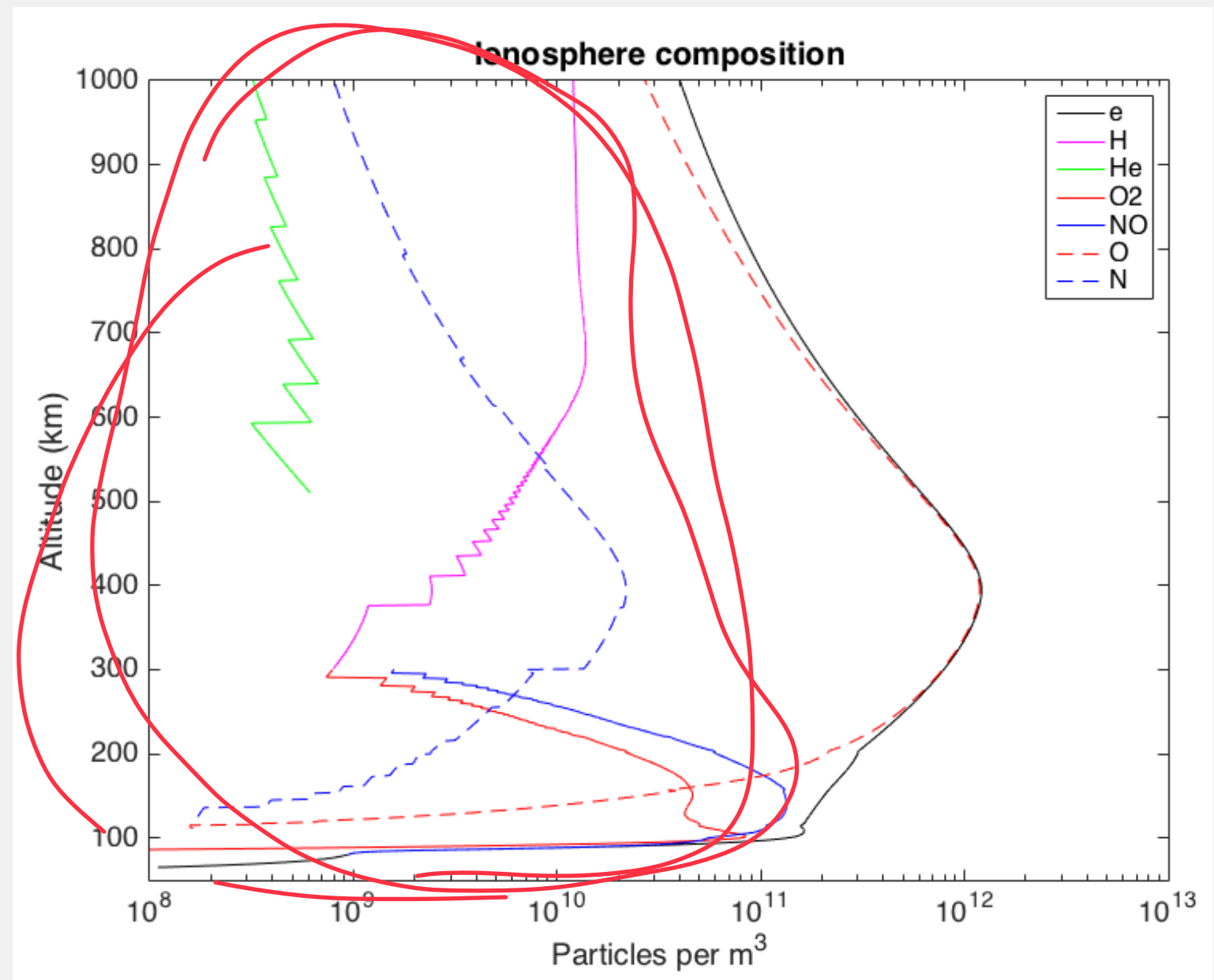
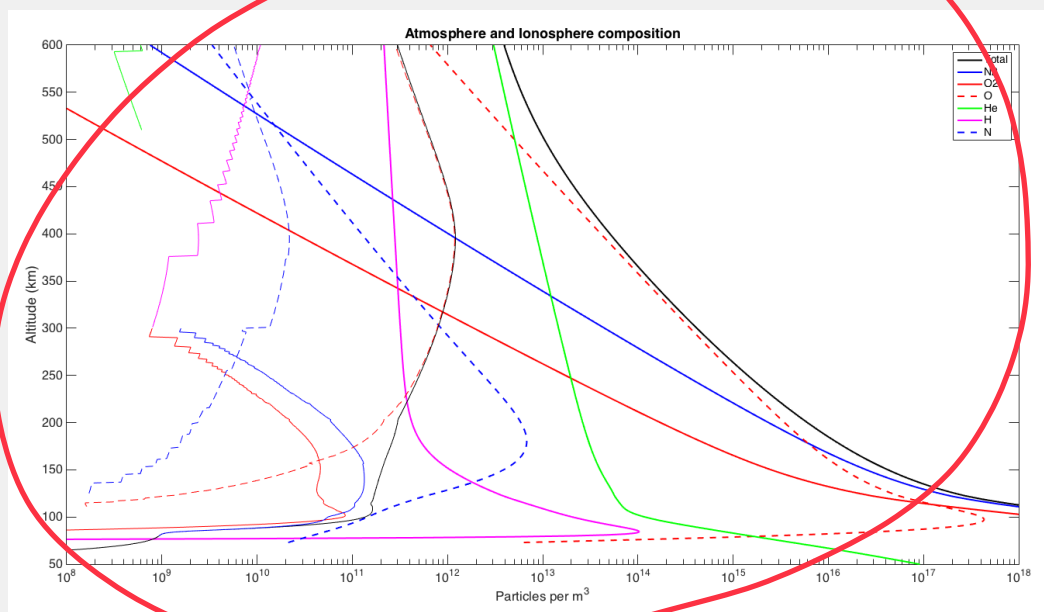
Sodankylä Ion and Neutral Chemistry (SIC) model:



Topside Ionosphere

- ❖ Above ~350 km, densities are so low that ions are not dominated by chemistry, but diffusive equilibrium

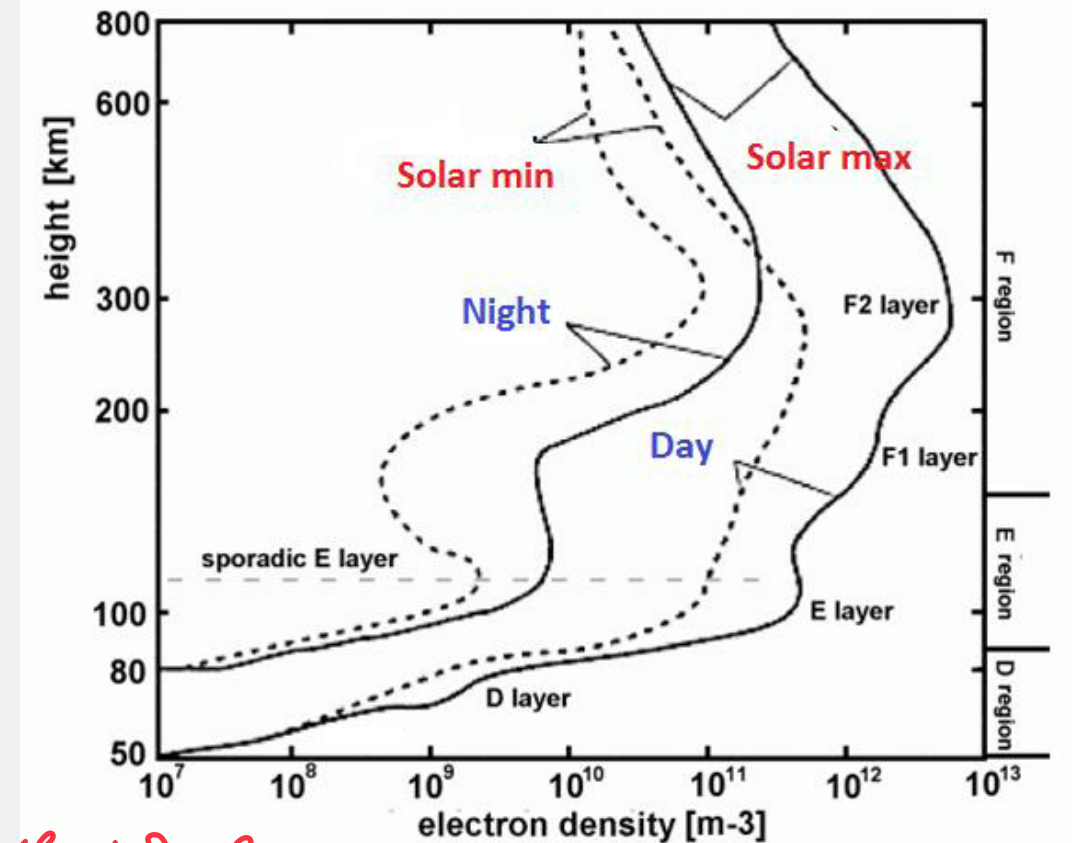
400 km



analogous to homosphere / heterosphere: bottom has turbulent mixing, top has diffusive equilibrium. In ionosphere case, lower altitudes have chemical equilibrium, higher has diffusive equilibrium.

Summary of Layers

- ❖ Layers are dominated by different ions
- ❖ Ion densities controlled by balance between production and loss; depends on chemical reactions in each altitude range
- ❖ Decay of layers at night depends on recombination rates and densities: low densities at high altitudes mean few collisions

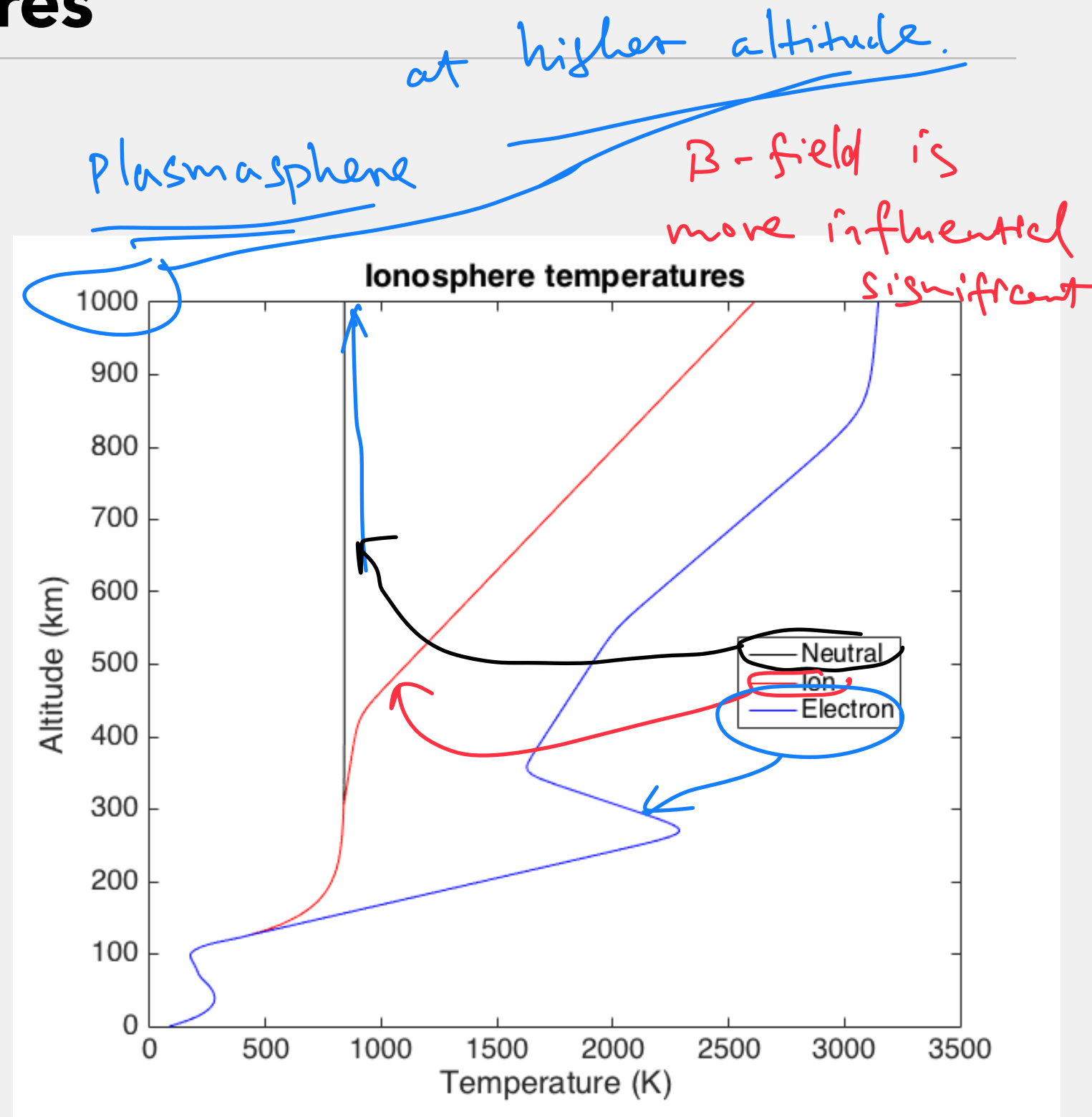


→ 50 km during very active time

Ionospheric Layer	Altitude Range (km)	Major Constituents	Notable Characteristics
D	70–90	NO ⁺ O ₂ ⁺ (molecular)	Disappears (recombines) very rapidly—minutes after sundown
E	90–140	O ₂ ⁺ (molecular) NO ⁺	Recombines rapidly—often disappears before midnight
F1	140–200	O ⁺ (atomic) NO ⁺	Mostly recombines after sundown, but pockets of ionization may remain
F2	200–400	O ⁺ (atomic)	Persistent because of low collision rates, but density decreases after sundown
Topside	> 400	O ⁺ (atomic) H ⁺	Merges into the plasmasphere, atomic oxygen dominates at lower altitudes, and hydrogen dominates at higher altitudes.

Ionosphere Temperatures

- ❖ Below 110 km, temperatures are made equal by collisions
- ❖ Above ~110 km, collisions are rare, so each species gets its temperature through different heating processes (radiation absorption, convection, etc)
- ❖ Above ~110 km: ions, electrons, and neutrals have different temperatures
 - ❖ Remember: this simply means they have different velocity / energy distributions



Ionosphere *V*ariations

Transport

- ❖ Variety of processes move plasma in the ionosphere:
 - ❖ **Winds**: neutral winds drag ions along with them, if collision frequency is high enough
 - ❖ **Drifts**: various forces cause plasma to “drift”: electric and magnetic fields; gravity; pressure; etc.
- ❖ **These contribute to complex spatial and temporal variations in the ionosphere**